

TwinWorld 2026 Challenge Final Verification Factsheet

Sparse-view 2D Gaussian Splatting with a corrected depth readout, and what the semantic metric actually rewards

An Kai Chen
National Tsing Hua University
kylechen.ankai@gmail.com

1. Team details

- Team name: kylechen1211
- Team leader: An Kai Chen, National Tsing Hua University, kylechen.ankai@gmail.com
- Rest of the team: none. This is a one-person entry.
- Team website: none.
- Codabench user name: kylechen1211. We kept the team name identical to the user name so the entry is unambiguous to match.
- Codabench ID of the final submission: 909861, final phase, submitted 2026-08-31 09:56 UTC, archive `submission_v0.30_bcc019.zip`.
- Final challenge score: 1.534. PSNR 19.95, SSIM 0.723, LPIPS 0.290, F-score 0.582, mIoU 3D 0.341; equivalently `rendering_score` 0.610, `geometry_score` 0.582, `semantic_score` 0.341.
- Source code, checkpoints and factsheet: <https://github.com/kylevirtuous1211/ECCV-TwinWorld-challenge>
- Complete results in the native representation: see the Google Drive link in the accompanying email (kept separate from the code link, as requested).

The geometry term is the mean of three thresholds and that mean is dominated by the loosest, which is worth making explicit. On the four TUM scenes that ship ground truth — the scorer computes F on TUM only — a development-phase archive of ours decomposes as F@5 cm 0.357, F@10 cm 0.521, F@20 cm 0.637 for a mean of 0.505. That is the released-seed cloud family and not the one in the final archive, so read it as the shape of the three-threshold mean rather than as a decomposition of the 0.582 above, which is a board figure over five withheld scenes we cannot decompose.

2. Method details

2.1. Overall pipeline

Fig. 1 is the whole method. It is 2D Gaussian Splatting [4] on gsplat [12] with no new network, no data from outside the release, and no pretrained weights anywhere in the path that produces a submission. Three changes account for most of the score, and each was measured on the six scenes that ship ground truth before it was adopted.

1. A dense MVS seed. The released train/sparse/0 reconstruction covers only 0.07–0.20% of pixels and is a crop of a larger model: every reprojection error in it is exactly zero and its image identifiers run in the 700s–800s inside a model of at least 819 images. Retriangulating densely from the released images with the released poses held fixed, then running COLMAP [10] dense MVS, replaces it. Worth +0.10 of F-score on the withheld scenes.

2. Reading depth at the ray–disc intersection. 2DGS represents a surface as oriented discs, so the depth a pixel should receive is the distance to where its ray crosses the disc it hit. gsplat’s native 2DGS path reports the centre depth of that disc for every pixel the splat covers, because depth is passed through the rasteriser as an extra colour channel indexed by Gaussian identifier. A disc tilted 60° spans 1.73 of its own radius in depth and is reported as a constant.

This is a known upstream defect and we do not claim it as a finding. It has been open since 2024-11-05 as gsplat issues #477 and #863, with PR #932 (“use ray-plane intersection depth instead of Gaussian center depth”) unmerged; the original 2DGS work and its reference rasteriser diff-surfel-rasterization compute the intersection correctly. What we contribute is its price on this benchmark. In a paired A/B on identical

checkpoints over the four TUM scenes that ship ground truth, reading it correctly moves `geometry_score` from 0.3888 to 0.4717, with precision and recall at 5 cm both rising and the clouds getting 7–11% smaller. Our `twins-world/raydepth.py` recovers the intersection from what an unmodified install already returns, so it needs no CUDA patch.

A consequence worth stating for the benchmark’s sake: any geometry baseline built on `gsplat`’s native 2DGS path rather than on `diff-surfel-rasterization` is measuring a different quantity from the one [4] defines, including through `gsplat`’s own normal-consistency loss.

3. Averaging three rendering seeds. Worth +0.026 of `rendering_score` against the single seed that would otherwise have shipped, and +0.021 against the best of the three — which is the fairer comparison and still the right choice, because the best seed can only be identified where truth exists, and truth exists on none of the scenes the final phase scores. Averaging needs no ground truth, so it applies to the seven withheld scenes exactly as it applies to the six released ones.

2.2. Extra data

No. Everything comes from the released `Twin-World_Datasets` (HuggingFace snapshot 8d5dfff) and from our own submissions’ returned scores. No external ground truth, no data outside the release, and no pretrained reconstruction of these scenes. The only pretrained weights involved anywhere are the VGG network inside the `lrips` package, used for local scoring of the LPIPS term and never in the path that produces a submission.

Two properties of how we use the released ground truth are disclosed in full in `DISCLOSURE.md` in the code bundle, and summarised in Sec. 4 below.

2.3. Training strategy

Two different models are trained per scene because the two terms want different things.

Geometry (9 TUM scenes $\times$ 5 seeds, and 4 Gold Coast scenes): 2DGS, 7000 iterations for TUM and 15,000 for Gold Coast, normal weight 0.05, distortion weight 0.01, intersection depth readout, initialised from the dense MVS seed. Seed 0 is fused into a point cloud; seeds 1–4 exist only to vote on it. The vote keeps a point when at least two of the other four placed a point within 5 cm. Measured on the four TUM scenes that ship ground truth this is worth +0.029 of `geometry_score`, 0.4730 to 0.5016. It also discards points: 49% on the released-seed clouds it was tuned on and 28.5% on the dense MVS clouds that shipped, which is

what lets the archive stay at the full 2 cm fusion voxel rather than thinning to fit the scorer’s point budget.

Rendering (13 scenes $\times$ 3 seeds): 2DGS, 7000 iterations, normal weight 0.05, distortion weight 0, same seed. The three renders are averaged per pixel.

Semantics: a random forest [9] (200 trees, depth 32, minimum leaf 2) over geometric features and the rendered colour, fitted on the reconstructed clouds rather than on ground truth. Fitting on the reconstruction and adding colour are together worth 0.15 of `semantic_score` in the corrected frame.

Frame correction: the released Gold Coast ground truth sits 1.1–1.4 m out of the camera frame its own images define. This is a property of the released data and not of our reconstruction — fitting the provided `points3D.ply` to the provided `3d_gt` recovers the same 0.25° rotation with nothing of ours in the loop. We measure it on the two Gold Coast scenes that ship ground truth, with the four TUM scenes that ship ground truth as the control, which read 0.014 to 0.074 m.

2.4. Complexity and cost

model family	Gaussians	time
TUM geometry, per seed	19.1 M	
Gold Coast geometry	28.7 M	
Rendering, per seed	25.5 M	43 min

Per scene the models run 1.4–2.5 M Gaussians for TUM geometry and 7.0–7.5 M for Gold Coast. A rendering model trains in 2.5–5.5 minutes. A full thirteen-scene rebuild is about six hours on one RTX 6000 Ada or about two hours as a Slurm array on H200 nodes; a single geometry job was sampled at 7 GB of 48 GB. No network is trained beyond the Gaussians themselves, so there is no parameter count in the usual sense; the submitted archive holds 71.7 M points across 13 clouds and 39 images.

2.5. Inference and submission

Nothing at submission time retrains. Geometry models are rendered to depth, fused, voted and thinned; Gold Coast models are labelled, shifted, and for `scene_011` and `scene_012` converted into the volume described in Sec. 4; rendering models are rendered and averaged. `NATIVE_TO_SUBMISSION.md` in the results bundle gives every command. Running it reproduces the two submitted Gold Coast clouds byte for byte, and all 39 renders in the bundle are byte-identical to those in the archive.

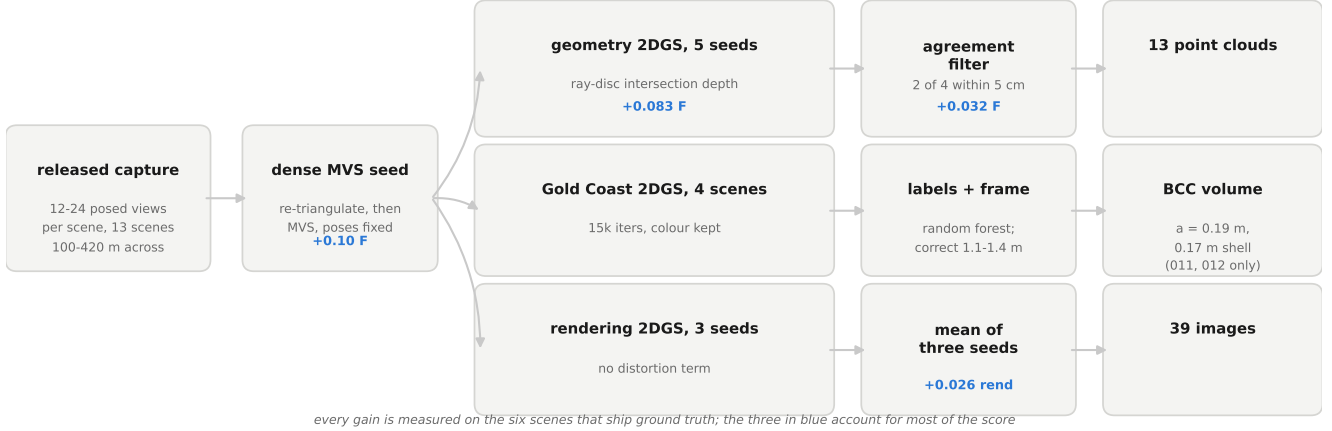


Figure 1. The pipeline. One dense seed feeds three separately trained families of Gaussians: geometry (five seeds, voted), Gold Coast geometry (which also carries the colour the label classifier consumes), and rendering (three seeds, averaged). Gains in blue are measured on the six scenes that ship ground truth.

2.6. Experimental results

Our row is Final 1.53, fourth of the twenty-one entries in the closed final phase. Within the challenge our F-score of 0.58 is close to the median of 0.57 and our mIoU of 0.34 is fourth against a median of 0.22, so the leaderboard separated entries on semantics rather than on geometry.

The internal ablations behind the three headline changes, each measured on the scenes that ship ground truth:

change	effect
centre → intersection depth	+0.083 F
sparse → dense MVS seed	+0.10 F (board)
one seed → 5-seed vote	+0.032 F
best seed → mean of three	+0.026 rend.
surface → labelled volume	+0.06 mIoU (board)

2.7. Novelty, and what is not ours

The depth readout is a known upstream defect, cited above; our contribution is the measurement, not the mechanism. The lattice result in Sec. 4 is Bambah [2] via Conway and Sloane [3], and Alam and Haas [1] had already applied optimal covering lattices to sensor coverage and derived the same 1.859 constant. That a one-sided metric can be gamed by thickening a surface is SparseOcc [7], which used the word “hacked” and measured +5–15 mIoU in occupancy prediction. Nothing in this submission has been previously published.

3. Ensemble and fusion strategies

Two ensembles are used and both are in the final submission.

Five-seed agreement filter (geometry). Five geometry models per TUM scene, differing only in seed. The cloud fused from seed 0 is filtered to the points that at least two of the other four seeds also placed within 5 cm. This is a precision filter rather than a fusion: it adds no points. It is worth +0.029 of geometry_score on the four scenes that ship truth, and it removes 28.5% of the points on the clouds that shipped. The second effect matters as much as the first, because the scorer rejects archives whose largest cloud is too big, so density and accuracy were otherwise in direct conflict.

Three-seed render averaging. The per-pixel mean of three rendering models. Worth +0.026 of rendering_score, and it beats the best of the three on five of the six scenes we can measure — so it is not only variance reduction; the seeds disagree mostly where each is separately wrong. The median across seeds is consistently worse than the mean (0.5979 against 0.6026), which is what a Gaussian-ish disagreement predicts.

We measured a third ensemble and rejected it: a floater prune gated on distance to the nearest training camera won on three scenes but lost over 18 paired A/Bs across six scenes and three seeds, mean −0.0012. It is not in the submission.

4. Disclosed properties of the submission

Stated here rather than left for the verification to find. All three are described at length in DISCLOSURE.md in the code bundle.

1. Released development-set ground truth is used

to place the test clouds. scene_009 and scene_010 ship ground truth so their frame offset is measured directly. scene_011 and scene_012 do not, and they are two of the seven scenes the final phase scores, so their offset is inferred — partly by carrying the offset measured on the other two, and partly by carrying scene_010’s ground-truth cloud into scene_011’s camera frame through a joint COLMAP reconstruction of the three registrable Gold Coast scenes, which overlap in a band. We read the terms’ “external ground-truth data” as data from outside the release, which this is not; we flag it because it is development-set ground truth influencing test-set predictions, and reverting it is a single parameter.

2. One of the three offset estimates uses our own returned scores. It models pooled mIoU as $ag(o - t)$, with g a response curve measured on the two scenes that ship truth, and fits t to five of our own scored submissions.

3. For scene_011 and scene_012 the submitted cloud is a labelled occupancy volume rather than a labelled surface. This optimises the semantic metric as written rather than improving the reconstruction, and it is the largest single component of our semantic score. It is discussed in Sec. 5.2, because we think the underlying observation is useful to the organisers independently of our entry.

5. Other details

5.1. Impressions, and difficulties encountered

The dataset is excellent and the three-term score is the right shape for a digital-twin task: it refuses to let a method win on appearance alone. The difficulty that dominated our effort was not the modelling. It was that the final phase scores seven scenes that ship no ground truth, so almost every choice had to be validated on six scenes and then transferred blind. Two of our worst mistakes came from that: we selected a Gold Coast representation on development clouds that had been thinned to 12 cm when the scenes it shipped to carry 6 cm, and a predicted factor of 2.26 arrived on the board as 1.29; and we shipped a scene whose reconstruction was missing a quarter of its extent for three submissions before a no-ground-truth diagnostic caught it.

A second difficulty worth reporting: the scorer rejects an archive above some point count with a null exit code and no log. We spent three submissions bracketing it. Publishing the limit, or returning any message at all on that path, would give every team those submissions back.

5.2. Comments on the evaluation metric

We think one property of the score is worth the organisers’ attention, and we report it as a measurement rather than an opinion.

The Gold Coast semantic term is a truth-driven nearest-neighbour lookup inside 10 cm with no precision component: a predicted point that no ground-truth point is near costs exactly nothing. On the two scenes that ship truth, deleting every false positive from our submission moves pooled mIoU from 0.3104 to 0.3207, while matching every ground-truth point with the labels we already assign would move it to 0.94 — a ratio of about 60:1 between coverage and precision.

Under that scoring the optimal submission is not a surface. It is a labelled occupancy volume, and the question becomes a covering problem: place N points so that as much ground truth as possible lies within 10 cm of one. We ship on the body-centred cubic lattice, the thinnest lattice covering of three-dimensional space [2, 3], which guarantees a 10 cm match for 349 points per cubic metre where the obvious cubic grid needs 649.

The part that surprised us is that thickness is worth nothing. We assumed a coarser lattice buying a thicker shell would win and measured it losing, monotonically in the covering radius. At a matched budget on scene_011 — 12.2 M against 12.0 M points — a bcc 0.25 lattice with a 0.45 m shell scores 0.342 where the shipped 0.15 m simple-cubic lattice scores 0.512; and across the whole sweep, at 8.7 to 12.2 M points, the order follows the covering radius and nothing else: 0.512, 0.342, 0.248, 0.172 for covering radii of 13.0, 14.0, 15.7 and 17.9 cm. Covered volume is nearly constant across those variants, 36,000 to 51,000 m³, while the score moves threefold — so volume is not the currency and position is. An oracle allowed to see the ground truth reaches the same score as our lattice with 17 times fewer points, which says the point budget was never the binding constraint: about 30% of the truth is simply not near anything we reconstructed.

Why this benchmark and not others. The same truth-driven nearest-neighbour label transfer is KITTI-360’s semantic mapping metric [6], and it is not exploitable there, because KITTI-360 reports geometric accuracy on the same points in the same table, so dilating the cloud destroys that column. ScanNet-family benchmarks cannot have the problem at all, because predictions are scored on a provided point set, so every prediction lands on a ground-truth point and false positives are counted automatically. Here geometry_score is computed on the TUM scenes and semantic_score on the Gold Coast scenes, and because those two sets are disjoint, nothing anywhere in the total penalises sub-

mitting volume where a surface belongs. We believe the decoupling, rather than the mIoU definition, is the thing worth changing.

5.3. Suggestions for future editions

1. Give the semantic term a precision component, evaluated on the same scenes — as [5] does for surfaces and as [6] does for this exact protocol. That removes the incentive described above entirely and costs one column.
2. Publish the point-count limit and return a message when an archive exceeds it. A null exit code with no log costs every team submissions to diagnose.
3. Consider whether 5 cm is measurable. TUM2TWIN’s own data-quality table [11, Tab. 2] gives the UAS laser scan 0.08 m absolute and 0.05 m relative accuracy, and UseGeo [8] reports a mean cloud-to-cloud error of 8.5 cm for conventional dense photogrammetry against airborne LiDAR with 224 images and at least eight rays per object point. On 12–24 views, F@5 cm may be partly a measurement of the reference rather than of the method. We state this as a question rather than a claim; we have not measured the reference ourselves.
4. Release ground truth for one withheld scene after the challenge closes. Every team is currently transferring choices blind from six scenes to seven, and the transfer is where our largest errors happened.
5. State whether development-set ground truth may inform test-set predictions. The Gold Coast georeferencing offset makes this decisive rather than academic, and the terms as written do not settle it.

5.4. A gap in what we can provide

The 39 rendering runs were launched without checkpoint saving, so the Gaussians behind the submitted images no longer exist. We checked whether retraining would recover them and it does not: two runs with byte-identical commands, the same seed and the same GPU end 31,224 Gaussians apart, with none of the three rendered frames byte-identical and a PSNR between the two runs of 15.7–23.8 dB. Training seeds only torch.manual_seed while gsplat’s backward accumulates with atomicAdd, whose summation order is not fixed. The results bundle therefore ships every rendered image from all three seeds, the 39 averaged ones that are byte-identical to the archive, and each run’s full configuration and metrics — but not retrained weights presented as the originals. Every submitted point cloud has its Gaussians except tum/scene_006, whose seed-0 run was launched without checkpoint saving for the same reason; its four voting seeds have theirs, and its

exported cloud, colour and receipt all ship.

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